

LAPORAN RESMI
PRAKTEK BASIS DATA LANJUT



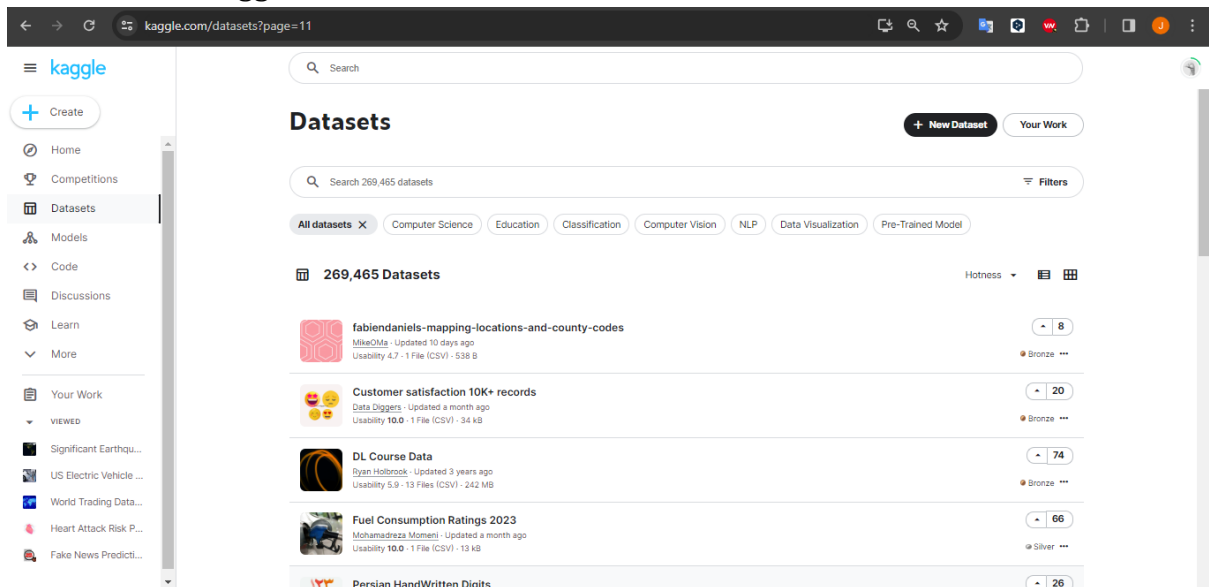
Dosen Pengampu
YUNIA IKAWATI S.ST, M.TR.KOM

Disusun Oleh
JUNDIAH INDIRA MILKA TSAURI
3122521020

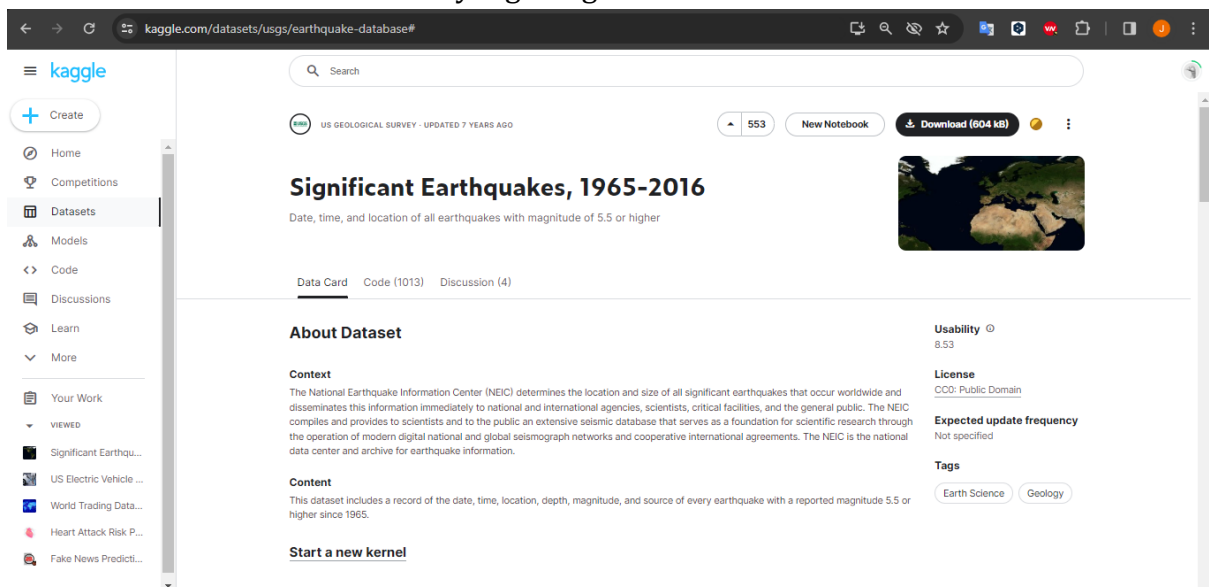
PROGRAM STUDI TEKNIK INFORMATIKA
DEPARTEMEN TEKNIK INFORMATIKA DAN KOMPUTER
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
2023

Langkah-langkah pembuatan:

1. Buka web kaggle.com untuk mencari dataset

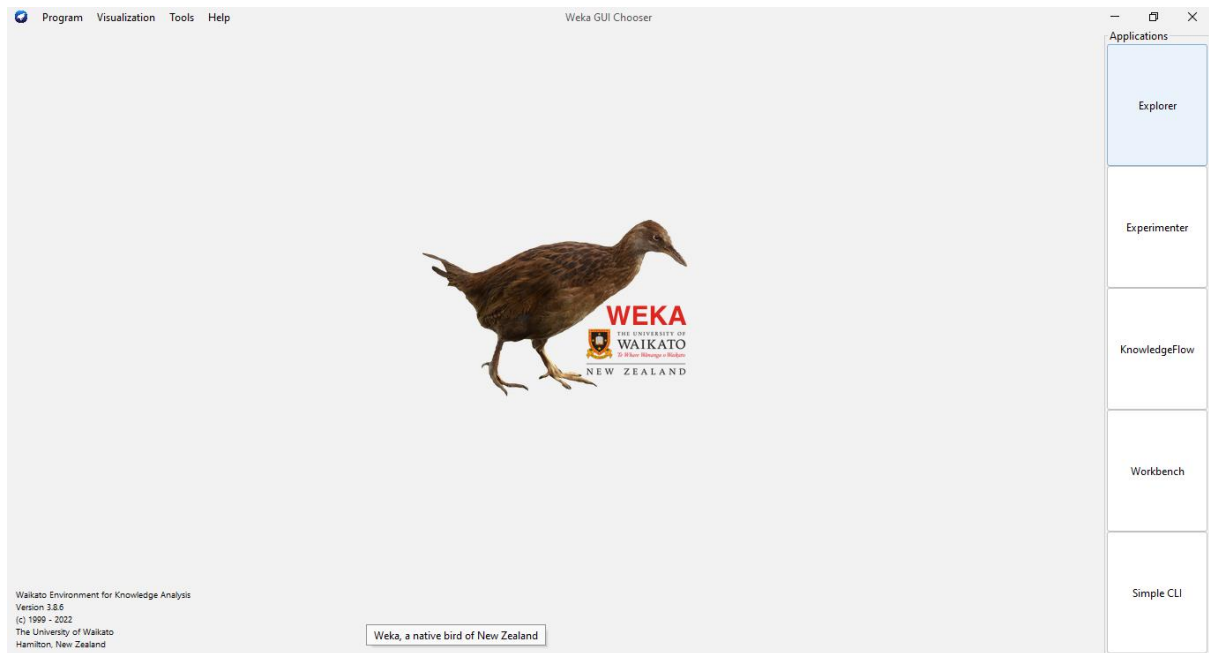


2. Setelah menemukan dataset yang diinginkan klik download

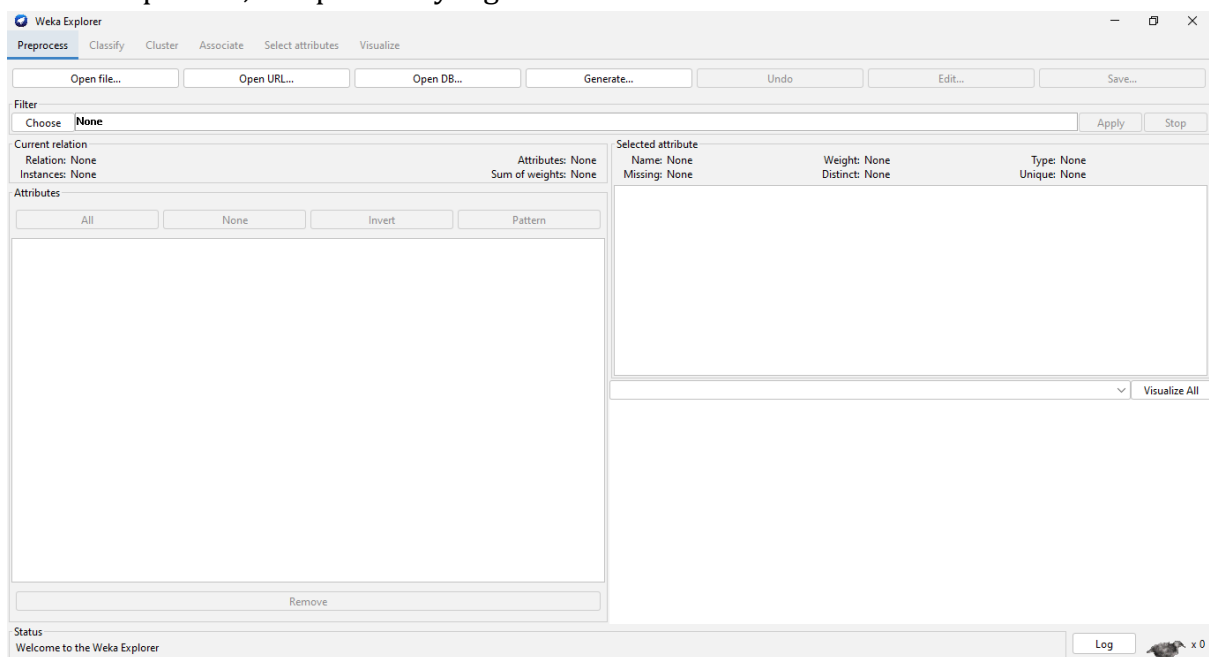


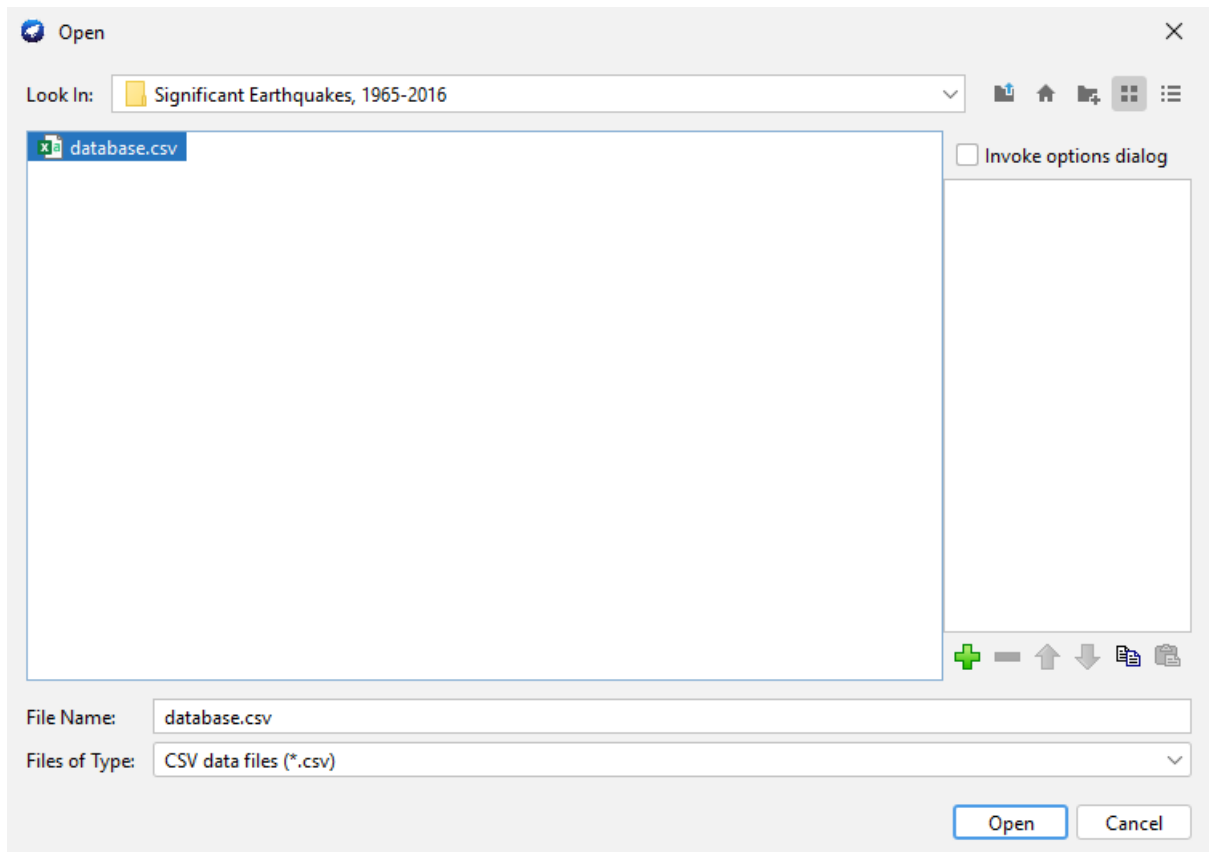
Setelah berhasil download, jika file berupa zip ekstrak terlebih dahulu.

3. Buka weka, pilih Explore

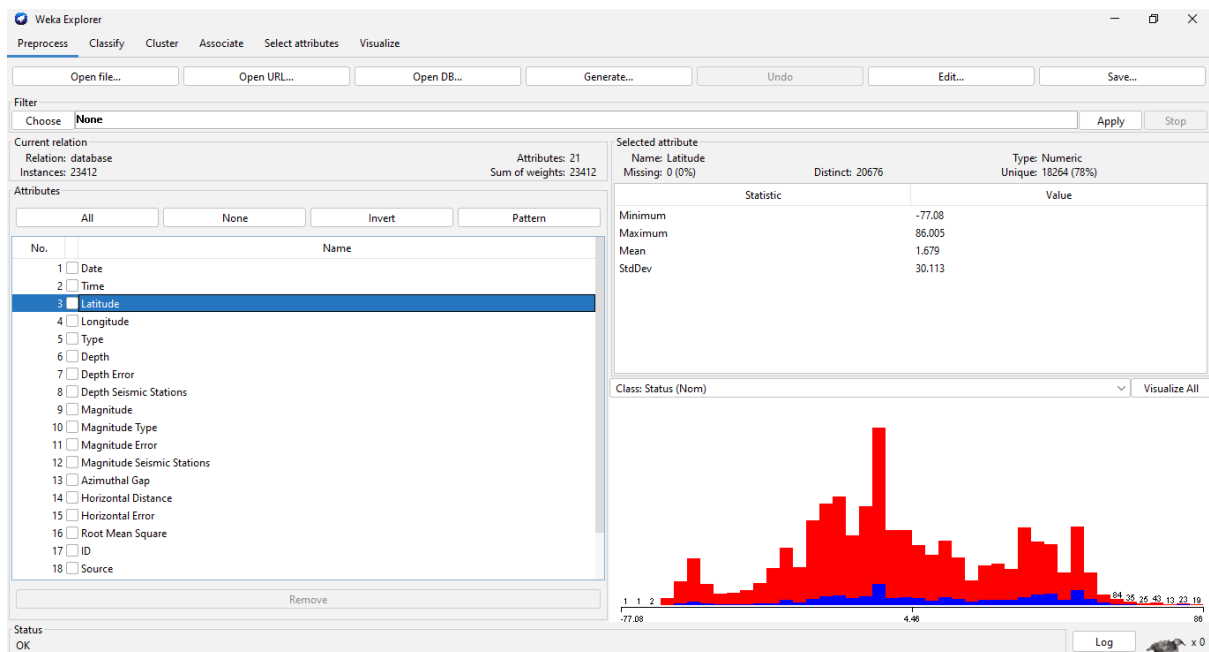


4. Klik open file, lalu pilih file yang akan diolah



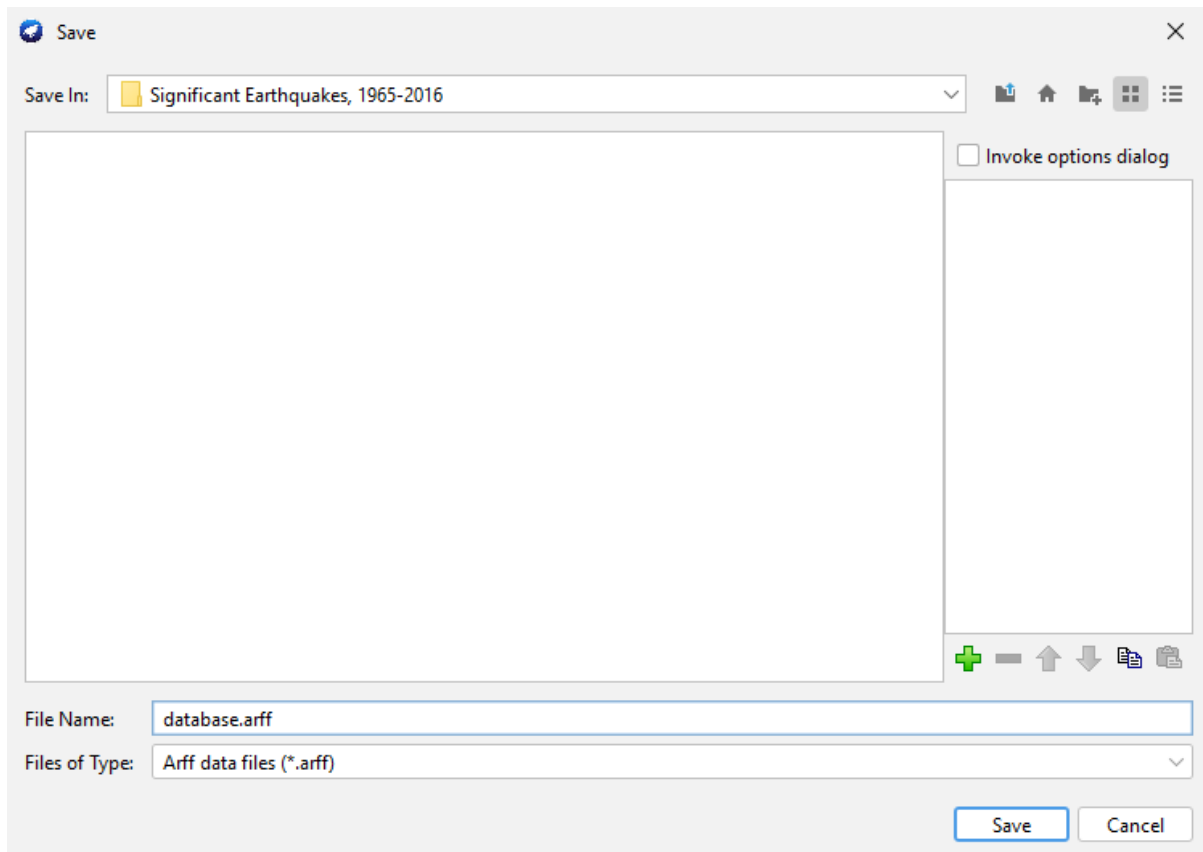


Pada langkah ini, file masih bertipe csv.

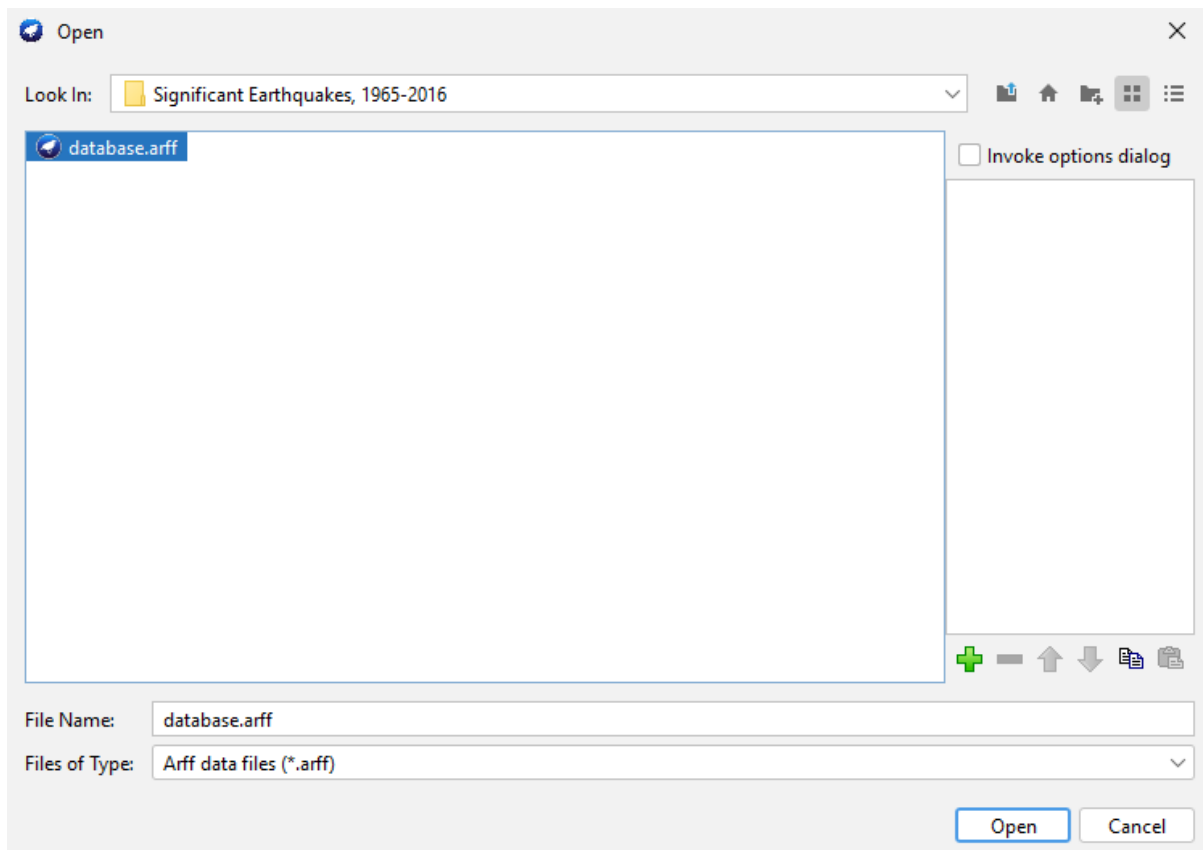


Ini adalah tampilan setelah file dibuka di weka.

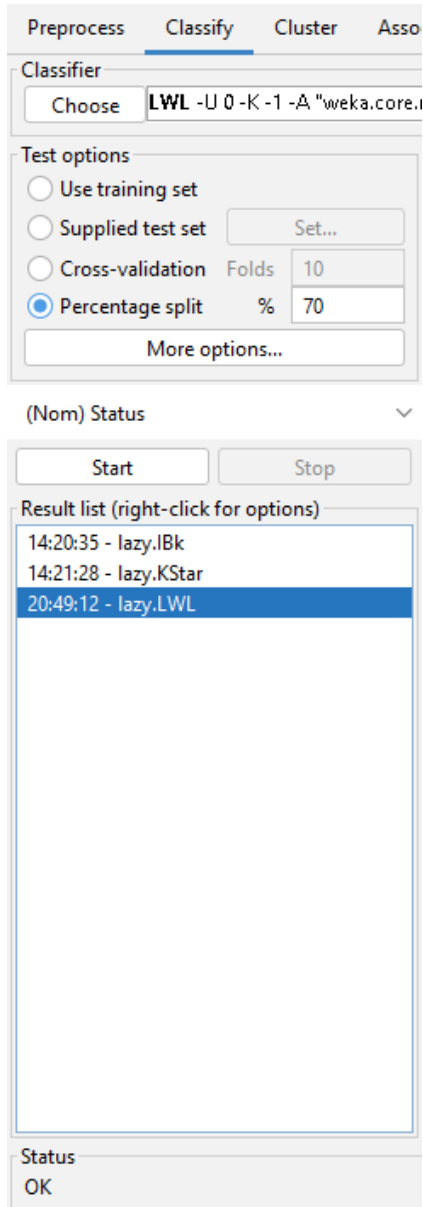
5. Klik save di pojok kanan atas, lalu ubah tipe file dari csv ke arff, lalu klik save



6. Pilih open file lagi, lalu pilih file yang diubah tadi jadi tipe arff, klik open



7. Pilih classify di kedua dari kiri atas, lalu pilih algoritma yang akan digunakan. Disini saya menggunakan tiga algoritma, yaitu lazy.IBk, lazy.KStar, lazy.LWL. Setelah memilih algoritma, klik Start



8. Berikut ini hasil pengolahan dataset menggunakan algoritma lazy IBk

```
Classifier output

=== Run information ===

Scheme:      weka.classifiers.lazy.IBk -K 1 -W 0 -A "weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance -R first-last\""
Relation:    database
Instances:   23412
Attributes:  21
             Date
             Time
             Latitude
             Longitude
             Type
             Depth
             Depth Error
             Depth Seismic Stations
             Magnitude
             Magnitude Type
             Magnitude Error
             Magnitude Seismic Stations
             Azimuthal Gap
             Horizontal Distance
             Horizontal Error
             Root Mean Square
             ID
             Source
             Location Source
             Magnitude Source
             Status
Test mode:   split 70.0% train, remainder test

=== Classifier model (full training set) ===

IB1 instance-based classifier
using 1 nearest neighbour(s) for classification

Time taken to build model: 0.04 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 25.26 seconds

=== Summary ===

Correctly Classified Instances      7024           100    %
Incorrectly Classified Instances      0             0    %
Kappa statistic                      1
Mean absolute error                  0.0001
Root mean squared error              0.0001
Relative absolute error              0.0305 %
Root relative squared error          0.0193 %
Total Number of Instances           7024

=== Detailed Accuracy By Class ===

              TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Automatic
1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000    Reviewed
Weighted Avg.    1.000    0.000    1.000    1.000    1.000    1.000    1.000    1.000

=== Confusion Matrix ===
      a    b  <-- classified as
792    0 |    a = Automatic
 0 6232 |    b = Reviewed
```

=== Run information ===

Scheme: weka.classifiers.lazy.IBk -K 1 -W 0 -A
"weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance
-R first-last\""
Relation: database
Instances: 23412
Attributes: 21
 Date
 Time
 Latitude
 Longitude
 Type
 Depth

Depth Error
Depth Seismic Stations
Magnitude
Magnitude Type
Magnitude Error
Magnitude Seismic Stations
Azimuthal Gap
Horizontal Distance
Horizontal Error
Root Mean Square
ID
Source
Location Source
Magnitude Source
Status

Test mode: split 70.0% train, remainder test

=== Classifier model (full training set) ===

IB1 instance-based classifier
using 1 nearest neighbour(s) for classification

Time taken to build model: 0.04 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 25.26 seconds

=== Summary ===

Correctly Classified Instances	7024	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.0001		
Root mean squared error	0.0001		
Relative absolute error	0.0305 %		
Root relative squared error	0.0193 %		
Total Number of Instances	7024		

=== Detailed Accuracy By Class ===

		TP Rate	FP Rate	Precision	Recall	F-Measure	MCC
ROC Area	PRC Area	Class					
1.000	1.000	Automatic	1.000	0.000	1.000	1.000	1.000

		1.000	0.000	1.000	1.000	1.000	1.000
1.000	1.000	Reviewed					
Weighted Avg.		1.000	0.000	1.000	1.000	1.000	1.000
1.000	1.000						

=== Confusion Matrix ===

```

a    b  <-- classified as
792   0 |    a = Automatic
  0 6232 |    b = Reviewed

```

9. Berikut hasil pengolahan dataset menggunakan algoritma lazy KStar

```

Classifier output
=== Run information ===

Scheme:      weka.classifiers.lazy.KStar -B 20 -M a
Relation:    database
Instances:   23412
Attributes:  21
    Date
    Time
    Latitude
    Longitude
    Type
    Depth
    Depth Error
    Depth Seismic Stations
    Magnitude
    Magnitude Type
    Magnitude Error
    Magnitude Seismic Stations
    Azimuthal Gap
    Horizontal Distance
    Horizontal Error
    Root Mean Square
    ID
    Source
    Location Source
    Magnitude Source
    Status
Test mode:   split 70.0% train, remainder test

=== Classifier model (full training set) ===

KStar Beta Verion (0.1b).
Copyright (c) 1995-97 by Len Trigg (trigg@cs.waikato.ac.nz).

Java port to Weka by Abdelaziz Mahoui (aml4@cs.waikato.ac.nz).

KStar options : -B 20 -M a

Time taken to build model: 0.02 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 386.8 seconds

=== Summary ===

Correctly Classified Instances      6982           99.4021 %
Incorrectly Classified Instances     42           0.5979 %
Kappa statistic                     0.9698
Mean absolute error                  0.006
Root mean squared error              0.0731
Relative absolute error              2.9998 %
Root relative squared error          23.1088 %
Total Number of Instances           7024

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
          0.963    0.002    0.983     0.963    0.973      0.970    0.993     0.978    Automatic
          0.998    0.037    0.995     0.998    0.997      0.970    0.994     0.998    Reviewed
Weighted Avg.   0.994    0.033    0.994     0.994    0.994      0.970    0.994     0.996

=== Confusion Matrix ===

  a    b  <-- classified as
 763   29 |    a = Automatic
 13 6219 |    b = Reviewed

```

=== Run information ===

Scheme: weka.classifiers.lazy.KStar -B 20 -M a
Relation: database
Instances: 23412
Attributes: 21
Date
Time
Latitude
Longitude
Type
Depth
Depth Error
Depth Seismic Stations
Magnitude
Magnitude Type
Magnitude Error
Magnitude Seismic Stations
Azimuthal Gap
Horizontal Distance
Horizontal Error
Root Mean Square
ID
Source
Location Source
Magnitude Source
Status
Test mode: split 70.0% train, remainder test

=== Classifier model (full training set) ===

KStar Beta Verion (0.1b).
Copyright (c) 1995-97 by Len Trigg (trigg@cs.waikato.ac.nz).
Java port to Weka by Abdelaziz Mahoui (am14@cs.waikato.ac.nz).

KStar options : -B 20 -M a

Time taken to build model: 0.02 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 386.8 seconds

=== Summary ===

Correctly Classified Instances	6982	99.4021 %
Incorrectly Classified Instances	42	0.5979 %
Kappa statistic	0.9698	
Mean absolute error	0.006	

Root mean squared error	0.0731
Relative absolute error	2.9998 %
Root relative squared error	23.1088 %
Total Number of Instances	7024

=== Detailed Accuracy By Class ===

		TP Rate	FP Rate	Precision	Recall	F-Measure	MCC
ROC Area	PRC Area	Class					
		0.963	0.002	0.983	0.963	0.973	0.970
0.993	0.978	Automatic					
		0.998	0.037	0.995	0.998	0.997	0.970
0.994	0.998	Reviewed					
Weighted Avg.		0.994	0.033	0.994	0.994	0.994	0.970
0.994	0.996						

=== Confusion Matrix ===

```

      a      b  <-- classified as
763   29 |    a = Automatic
13 6219 |    b = Reviewed

```

10. Berikut hasil pengolahan dataset menggunakan algoritma lazy LWL

```

Classifier output
=== Run information ===

Scheme:      weka.classifiers.lazy.LWL -U 0 -K -1 -A "weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance -R first-last\"" -W weka.cl
Relation:    database
Instances:    23412
Attributes:   21
Date
Time
Latitude
Longitude
Type
Depth
Depth Error
Depth Seismic Stations
Magnitude
Magnitude Type
Magnitude Error
Magnitude Seismic Stations
Azimuthal Gap
Horizontal Distance
Horizontal Error
Root Mean Square
ID
Source
Location Source
Magnitude Source
Status
Test mode:   split 70.0% train, remainder test

```

=== Classifier model (full training set) ===

Locally weighted learning

=====

Using classifier: weka.classifiers.trees.DecisionStump

Using linear weighting kernels

Using all neighbours

Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 499.3 seconds

=== Summary ===

Correctly Classified Instances	7024	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.0009		
Root mean squared error	0.0044		
Relative absolute error	0.4272 %		
Root relative squared error	1.3859 %		
Total Number of Instances	7024		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Automatic
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Reviewed
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a	b	<-- classified as
792	0	a = Automatic
0	6232	b = Reviewed

=== Run information ===

Scheme: weka.classifiers.lazy.LWL -U 0 -K -1 -A

"weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance
-R first-last\" -W weka.classifiers.trees.DecisionStump

Relation: database

Instances: 23412

Attributes: 21

Date

Time

Latitude

Longitude

Type

Depth

Depth Error

Depth Seismic Stations

Magnitude

Magnitude Type

Magnitude Error

Magnitude Seismic Stations

Azimuthal Gap

Horizontal Distance

Horizontal Error

Root Mean Square

ID

Source

Location Source

```

          Magnitude Source
          Status
Test mode:  split 70.0% train, remainder test

```

=== Classifier model (full training set) ===

Locally weighted learning

=====

Using classifier: weka.classifiers.trees.DecisionStump

Using linear weighting kernels

Using all neighbours

Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 499.3 seconds

=== Summary ===

Correctly Classified Instances	7024	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.0009		
Root mean squared error	0.0044		
Relative absolute error	0.4272 %		
Root relative squared error	1.3859 %		
Total Number of Instances	7024		

=== Detailed Accuracy By Class ===

ROC Area	PRC Area	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC
		Class					
		1.000	0.000	1.000	1.000	1.000	1.000
1.000	1.000	Automatic					
		1.000	0.000	1.000	1.000	1.000	1.000
1.000	1.000	Reviewed					
Weighted Avg.		1.000	0.000	1.000	1.000	1.000	1.000
1.000	1.000						

=== Confusion Matrix ===

```

      a      b  <-- classified as
792    0 |    a = Automatic
    0 6232 |    b = Reviewed

```

11. Untuk melihat visualisasinya saya menambah algoritma lagi, yaitu trees.J48

The screenshot shows the Weka Classifier window. The 'Classifier' tab is active, and 'J48 -C 0.25 -M 2' is selected in the 'Choose' dropdown. Under 'Test options', 'Percentage split' is selected with a value of 80%. The 'Result list' shows four entries: '11:51:31 - lazy.IBk', '11:52:59 - lazy.KStar', '11:59:33 - lazy.LWL', and '14:32:53 - trees.J48', with the last one highlighted. The 'Start' button is visible.

Berikut hasil pengolahan dataset dengan trees J48

```
Classifier output

=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    database
Instances:    23412
Attributes:   21
              Date
              Time
              Latitude
              Longitude
              Type
              Depth
              Depth Error
              Depth Seismic Stations
              Magnitude
              Magnitude Type
              Magnitude Error
              Magnitude Seismic Stations
              Azimuthal Gap
              Horizontal Distance
              Horizontal Error
              Root Mean Square
              ID
              Source
              Location Source
              Magnitude Source
              Status
Test mode:    split 80.0% train, remainder test

=== Classifier model (full training set) ===

J48 pruned tree
-----

Source = ISOGEM: Automatic (2460.0)
Source = ISOGEMSUP: Automatic (120.0)
Source = OFFICIAL: Reviewed (8.0/1.0)
Source = CI: Reviewed (61.0)
Source = US: Reviewed (20630.0)
Source = NC: Reviewed (51.0)
Source = GCHT: Automatic (55.0)
Source = UW: Reviewed (6.0)
Source = ATLAS: Automatic (3.0)
Source = MN: Reviewed (4.0)
Source = SE: Reviewed (1.0)
Source = AK: Reviewed (12.0)
Source = PR: Reviewed (1.0)

Number of Leaves :    13

Size of the tree :    14

Time taken to build model: 0.47 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.06 seconds
```

```

=== Summary ===

Correctly Classified Instances      4682          100   %
Incorrectly Classified Instances      0           0   %
Kappa statistic                      1
Mean absolute error                  0.0001
Root mean squared error              0.0038
Relative absolute error              0.0477 %
Root relative squared error          1.2093 %
Total Number of Instances           4682

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
1.000   0.000   1.000     1.000   1.000     1.000   1.000   1.000   Automatic
1.000   0.000   1.000     1.000   1.000     1.000   1.000   1.000   Reviewed
Weighted Avg.   1.000   0.000   1.000     1.000   1.000     1.000   1.000   1.000

=== Confusion Matrix ===
      a    b  <-- classified as
526    0 |  a = Automatic
 0 4156 |  b = Reviewed

```

=== Run information ===

```

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    database
Instances:   23412
Attributes:  21
              Date
              Time
              Latitude
              Longitude
              Type
              Depth
              Depth Error
              Depth Seismic Stations
              Magnitude
              Magnitude Type
              Magnitude Error
              Magnitude Seismic Stations
              Azimuthal Gap
              Horizontal Distance
              Horizontal Error
              Root Mean Square
              ID
              Source
              Location Source
              Magnitude Source
              Status
Test mode:   split 80.0% train, remainder test

```

=== Classifier model (full training set) ===

J48 pruned tree

Source = ISCGEM: Automatic (2460.0)
Source = ISCGEMSUP: Automatic (120.0)
Source = OFFICIAL: Reviewed (8.0/1.0)
Source = CI: Reviewed (61.0)
Source = US: Reviewed (20630.0)
Source = NC: Reviewed (51.0)
Source = GCMT: Automatic (55.0)
Source = UW: Reviewed (6.0)
Source = ATLAS: Automatic (3.0)
Source = NN: Reviewed (4.0)
Source = SE: Reviewed (1.0)
Source = AK: Reviewed (12.0)
Source = PR: Reviewed (1.0)

Number of Leaves : 13

Size of the tree : 14

Time taken to build model: 0.47 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.06 seconds

=== Summary ===

Correctly Classified Instances	4682	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.0001		
Root mean squared error	0.0038		
Relative absolute error	0.0477 %		
Root relative squared error	1.2093 %		
Total Number of Instances	4682		

=== Detailed Accuracy By Class ===

		TP Rate	FP Rate	Precision	Recall	F-Measure	MCC
ROC Area	PRC Area	Class					
1.000	1.000	1.000 Automatic	0.000	1.000	1.000	1.000	1.000
1.000	1.000	1.000 Reviewed	0.000	1.000	1.000	1.000	1.000
Weighted Avg.		1.000	0.000	1.000	1.000	1.000	1.000
1.000	1.000						

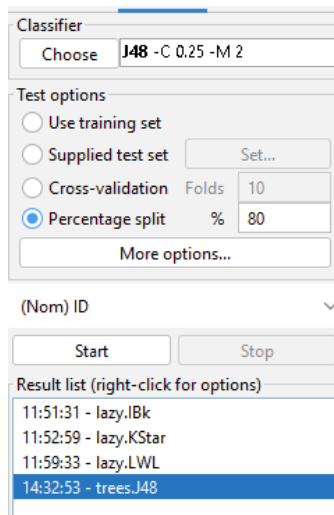
=== Confusion Matrix ===

```

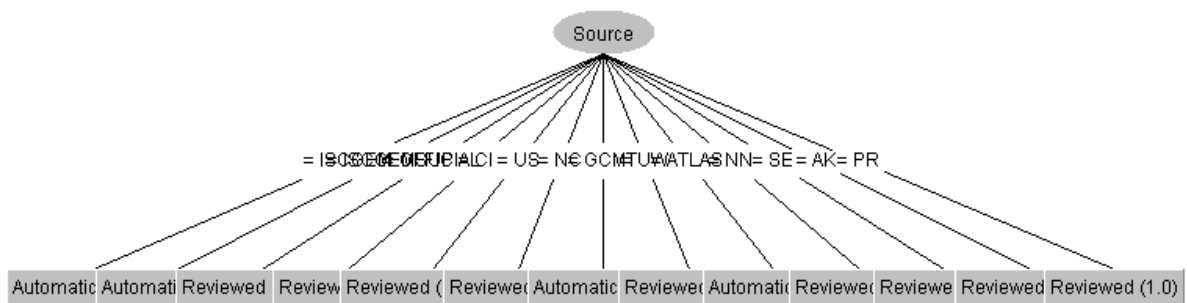
a    b  <-- classified as
526   0 |    a = Automatic
  0 4156 |    b = Reviewed

```

Untuk menampilkan visualize tree nya, klik kanan pada 14:32:53 – trees.J48 lalu pilih visualize tree



Berikut hasil dari visualize tree nya



Analisa

Dataset tersebut berisi tentang data tanggal, waktu, dan lokasi semua gempa bumi dengan magnitudo 5,5 atau lebih tinggi dari tahun 1965-2016. Saya mengolah dataset menggunakan aplikasi WEKA versi 3.8.6. Algoritma yang saya gunakan ada empat yaitu lazy-IBk, lazy-KStar, lazy-LWL, dan trees-J48. Dataset ini memiliki relation berupa database, 23412 intances, 21 atribut.

Untuk lazy-IBk dan lazy-LWL memiliki hasil pengolahan data yang sama, namun untuk pengambilan waktunya berbeda, lazy-LWL lebih cepat untuk membangun model tapi lebih lama ketika mengetes modelnya. Perbedaan hasil pengolahan data pada algoritma lazy-IBk dan lazy-KStar dapat dilihat pada summary (correctly dan incorrectly classified instances, kappa statistic, mean absolute error, root mean squared error, relative absolute error, root relative squared error), detailed accuracy by class, dan confusion matrix. Waktu untuk membangun model dan mengetes model juga berbeda. Lazy-IBk memiliki waktu membangun model lebih lama dibandingkan lazy-KStar, tapi untuk mengetes model lazy-IBk lebih cepat dibanding lazy-KStar.

Perbedaan utama untuk algoritma lazy-IBk dan lazy-KStar ada pada yang pertama jenis klasifikasi, IBk menggunakan klasifikasi tetangga terdekat (k-Nearest Neighbors) sedangkan KStar juga klasifikasi tetangga terdekat tapi dengan opsi yang berbeda. Kedua opsi klasifikasi, IBk menggunakan 1 tetangga dekat (K 1) sedangkan KStar menggunakan 20 tetangga (-B 20), pengaturan yang berbeda untuk penanganan keadaan seimbang dan pemecahan keadaan seimbang (opsi -W 0 dan -M a). Ketiga metrik kinerja, hasil pengolahan dengan IBk dan KStar mencapai akurasi tinggi, tapi KStar memiliki sedikit kesalahan klasifikasi, statistika kappa mendekati 1 pada kedua hasil pengolahan yang mana menunjukkan kesepakatan yang tinggi. Keempat akurasi rinci berdasarkan kelas, IBk memiliki presisi, recall, dan f-measure 100% untuk kedua kelas sedangkan KStar memiliki metrik yang sedikit lebih rendah, tapi masih sangat tinggi. Secara keseluruhan, IBk dan KStar memiliki kinerja yang sangat baik, tetapi terdapat perbedaan dalam opsi, jumlah tetangga, dan metrik rinci. Pilihan antara IBk dan KStar ini mungkin tergantung pada persyaratan khusus dan pertimbangan untuk dataset dan masalah yang diberikan.

Perbedaan utama untuk algoritma lazy-LWL dan trees-J48, pertama metode klasifikasi, J48 dengan pruning menciptakan pohon keputusan terstruktur sedangkan LWL dengan decision stump yang memberikan bobot linier ke tetangga. Kedua bentuk

model, J48 menghasilkan pohon keputusan terstruktur dengan 13 daun sedangkan LWL menggunakan decision stump tanpa pembagian sehingga hanya memiliki 1 daun. Ketiga waktu pemrosesan, J48 membutuhkan waktu lebih sedikit untuk membangun model dan pengujian model, sedangkan LWL membutuhkan waktu lebih lama untuk pengujian model, membangun model tidak diukur karena menggunakan model instance-based. Keempat metrik performa, keduanya memiliki performa sempurna dengan akurasi 100% dan statistik kappa 1.